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IIT-JAM

MATHEMATICS - 2005

Objective Questions : (Q.1 – Q.15)

- (1.) Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$.

Then $\{a_n\}$ is convergent

- (a.) Implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent
(b.) Implies $\{c_n\}$ is convergent but $\{b_n\}$ need not be convergent
(c.) Implies both $\{b_n\}$ and $\{c_n\}$ are convergent
(d.) If both $\{b_n\}$ and $\{c_n\}$ are convergent

- (2.) An integrating factor of

$$x \frac{dy}{dx} + (3x + 1)y = xe^{-2x} \text{ is}$$

- (a.) xe^{3x}
(b.) $3xe^x$
(c.) xe^x
(d.) x^3e^x

(3.) The general solution of $x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 9y = 0$ is

(a.) $(c_1 + c_2 x) e^{3x}$

(b.) $(c_1 + c_2 \ln x) x^3$

(c.) $(c_1 + c_2 x) x^3$

(d.) $(c_1 + c_2 \ln x) e^{x^3}$

(Here c_1 and c_2 are arbitrary constants)

(4.) Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. If $\phi(x, y, z)$ is a solution of the Laplace equation then the vector field $(\vec{\nabla}\phi + \vec{r})$ is

(a.) Neither solenoidal nor irrotational

(b.) Solenoidal but not irrotational

(c.) Both solenoidal and irrotational

(d.) Irrotational but not solenoidal

(5.) Let $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$, S be the surface of the sphere $x^2 + y^2 + z^2 = 1$ and $\hat{n}$ be the inward unit normal vector to S . Then $\iint_S \vec{F} \cdot \hat{n} dS$ is

equal to

(a.) 4π

(b.) -4π

(c.) 8π

(d.) -8π

- (6.) Let A be a 3×3 matrix with eigenvalues 1, -1 and 3. Then
- (a.) $A^2 + A$ is non-singular
 - (b.) $A^2 - A$ is non-singular
 - (c.) $A^2 + 3A$ is non-singular
 - (d.) $A^2 - 3A$ is non-singular
- (7.) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation and I be the identity transformation of $\mathbb{R}^3$. If there is a scalar c and a non-zero vector $x \in \mathbb{R}^3$ such that $T(x) = cx$, then rank of $(T - cI)$
- (a.) Cannot be 0
 - (b.) Cannot be 1
 - (c.) Cannot be 2
 - (d.) Cannot be 3
- (8.) In the group $\{1, 2, \dots, 16\}$ under the operation of multiplication modulo 17, the order of the element 3 is
- (a.) 4
 - (b.) 8
 - (c.) 12
 - (d.) 16
- (9.) A ring R has maximal ideals
- (a.) If R is infinite
 - (b.) If R is finite

(c.) If R is finite with at least 2 elements

(d.) Only if R is finite

(10.) The integral $\int_0^1 \left[\int_0^{1-z} \left(\int_0^2 dx \right) dy \right] dz$ is equal to

(a.) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^2 dx \right) dz \right] dy$

(b.) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^1 dx \right) dz \right] dy$

(c.) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-z} dx \right) dz \right] dy$

(d.) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-y} dx \right) dz \right] dy$

(11.) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be continuous and $g, h : \mathbf{R}^2 \rightarrow \mathbf{R}$ be differentiable. Let

$F(u, v) = \int_v^u f(t) dt$, where $u = g(x, y)$ and

$v = h(x, y)$. Then $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} =$

(a.) $f(g(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(h(x, y)) \left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \right]$

(b.) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

(c.) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

$$(d.) f(g(x, y)) \left[\frac{\partial g}{\partial x} - \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$$

(12.) Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a, b]$.

Let L = length of the curve between $x = a$ and $x = b$.

A = area bounded by the curve, x -axis, and the lines $x = a$ and $x = b$.

S = area of the surface generated by revolving the curve about x -axis between $x = a$ and $x = b$.

Then

(a.) $2\pi KL < S < 2\pi A$

(b.) $S \leq 2\pi A < 2\pi KL$

(c.) $2\pi A \leq S < 2\pi KL$

(d.) $2\pi A < 2\pi KL < S$

(13.) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(t) = t^2$ and let

U be any non-empty open subset of $\mathbf{R}$. Then,

(a.) $f(U)$ is open

(b.) $f^{-1}(U)$ is open

(c.) $f(U)$ is closed

(d.) $f^{-1}(U)$ is closed

(14.) Let $f : (-1, 1) \rightarrow \mathbf{R}$ be such that $f^{(n)}(x)$ exists and $|f^{(n)}(x)| \leq 1$ for every $n \geq 1$ and for every $x \in (-1, 1)$. Then f has a convergent power series expansion in a neighbourhood of

- (a.) Every $x \in (-1, 1)$
- (b.) Every $x \in \left(-\frac{1}{2}, 0\right)$ only
- (c.) No $x \in (-1, 1)$
- (d.) Every $x \in \left(0, \frac{1}{2}\right)$ only

(15.) Let $a > 1$ and $f, g, h : [-a, a] \rightarrow \mathbf{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$.

$f(x) = 0$ only for $x = -a, 0, a$;

$f'(x) = 0 = g(x)$ only $x = -1, 0, 1$;

$g'(x) = 0 = h(x)$ only for $x = -c, c$.

The possible relations between f, g, h are

- (a.) $f = g'$ and $h = f'$
- (b.) $f' = g$ and $g' = h$
- (c.) $f = -g'$ and $h' = g$
- (d.) $f = -g'$ and $h' = f$

Subjective Questions: (Q.16-Q.22)

(16.) (a.) Solve the initial value problem

$$\frac{d^2 y}{dx^2} - y = x(\sin x + e^x), \quad y(0) = y'(0) = 1$$

(b.) Solve the differential equation

$$(2y \sin x + 3y^4 \sin x \cos x) dx - (4y^3 \cos^2 x + \cos x) dy = 0$$

(17.) Let G be a finite abelian group of order n with identity e . If for all $a \in G, a^3 = e$, then, by induction on n , show that $n = 3^k$ for some non-negative integer k .

(18.) (a.) Let $f : [a, b] \rightarrow \mathbf{R}$ be a differentiable function. Show that there exist point $c_1, c_2 \in (a, b)$ such that $2f(c_1)f'(c_1) = f'(c_2)[f(a) + f(b)]$.

(b.) Let

$$f(x, y) = \begin{cases} \frac{(x^2 + y^2)}{[\ln(x^2 + y^2) + 1]} & \text{for } (x, y) \neq (0, 0) \\ \alpha & \text{for } (x, y) = (0, 0) \end{cases}$$

Find a suitable value for α such that f is continuous. For this value of α is f differentiable at $(0, 0)$? Justify your claim.

(19.) (a.) Let S be the surface $x^2 + y^2 + z^2 = 1$, $z \geq 0$.

Use Stoke's theorem to evaluate

$$\int_C [(2x - y) dx - y dy - z dz]$$

where C is the circle $x^2 + y^2 = 1$, $z = 0$, oriented anticlockwise.

(b.) Show that the vector field

$$\vec{F} = (2xy - y^4 + 3)\hat{i} + (x^2 - 4xy^3)\hat{j}$$

is conservative. Find its potential and also the work done in moving a particle from $(1, 0)$ to $(2, 1)$ along some curve.

(20.) Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined by $T(x, y, z)$

$= (y + z, z, 0)$. Show that T is a linear

transformation. If $v \in \mathbf{R}^3$ is such that $T^2(v) \neq 0$, then show that

$B = \{v, T(v), T^2(v)\}$ forms a basis of $\mathbf{R}^3$.

Compute the matrix of T with respect to B .

Also find a $v \in \mathbf{R}^3$ such that $T^2(v) \neq 0$.

(21.) (a.) For each $n \in \mathbf{N}$, define $f_n : [-1, 1] \rightarrow \mathbf{R}$ by

$$f_n(x) = \begin{cases} 4n^2x & \text{for } x \in \left[0, \frac{1}{2n}\right] \\ -4n^2\left(x - \frac{1}{n}\right) & \text{for } x \in \left[\frac{1}{2n}, \frac{1}{n}\right] \\ 0 & \text{for } x \in \left[\frac{1}{n}, 1\right] \end{cases}$$

Compute $\int_0^1 f_n(x) dx$ for each n . Analyse pointwise and uniform convergence of the sequence of function $\{f_n\}$.

(b.) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a continuous function with $|f(x) - f(y)| \geq |x - y|$ for every $x, y \in \mathbf{R}$. Is f one-one? Show that there cannot exist three points $a, b, c \in \mathbf{R}$ with $a < b < c$ such that $f(a) < f(c) < f(b)$.

(22.) Find the volume of the cylinder with base as the disk of unit radius in the xy -plane centred at $(1, 1, 0)$ and the top being the surface

$$z = \left[(x-1)^2 + (y-1)^2 \right]^{3/2}.$$

Rajendra Dubey (Known as Dubey Sir): He has experience of almost three decades of teaching mathematics. During his journey, he has guided IIT-JEE, IAS, NET, GATE, IIT-JAM aspirants etc. Apart from producing top rankers in the various exams, he is widely known for making mathematics interesting and fascinating. He is founder & director of DIPS ACADEMY and chairman of Dubey's Group of Companies.



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